

- 30.** The Sun temple of Konark was built by Narasimhadeva I. To which dynasty did he belong to?  
 (a) Somavamsi dynasty  
 (b) Imperial Ganga dynasty  
 (c) Suryavanshi Gajapati dynasty  
 (d) Bhoi dynasty
- 31.** Whose philosophy is called the Advaita?  
 (a) Ramanujacharya (b) Shankaracharya  
 (c) Nagarjuna (d) Vasumitra
- 32.** Which one of the following was the original name of Tansen, the famous musician in the court of Akbar?  
 (a) Mahananda Pande  
 (b) Lal Kalwant  
 (c) Baz Bahadur  
 (d) Ramtanu Pandey

- 33.** Which one of the following dances involves solo performance?  
 (a) Bharatanatyam (b) Kuchipudi  
 (c) Kathak (d) Odissi
- 34.** Who among the following was a Jahangiri painter?  
 (a) Abul Hasan (b) Abdus Samad  
 (c) Daswant (d) Mir Sayyid Ali
- 35.** Consider the following statement(s) regarding Sanchi Stupas  
 1. Sanchi Stupa is located in Sanchi, Madhya Pradesh, 14 km from Vidisha.  
 2. It has three Stupas all the gateway around them but the most famous is the great Stupa which was originally made of brick in Ashoka's time.

Which of the statements(s) given above is/are correct?  
 (a) Only 1 (b) Only 2  
 (c) Both 1 and 2 (d) Neither 1 nor 2

- 36.** Which one of the following is an Octagonal Tomb?  
 (a) Tomb of Sikander Lodi  
 (b) Tomb of Balban  
 (c) Tomb of Ghiyasuddin Tughlaq  
 (d) Tomb of Ferozshah Tughlaq
- 37.** Which of the following tomb is placed in the centre of a large garden and resembles as a prototype of the Taj Mahal?  
 (a) Akbar's tomb at Sikandara  
 (b) Itmad-ud-daulah tomb at Agra  
 (c) Shershah's tomb at Sasaram  
 (d) Humayun's tomb at Delhi

## QUESTIONS FROM CDS EXAM (2012-2016)

### 2012 (I)

- 1.** The Viceregal Lodge at Shimla is a well-known ancient monument. Which of the following statements about the monument is/are correct?  
 1. The Lodge was built by 17th Viceroy, Earl Dufferin.  
 2. The present shape of the building was given by Earl of Marquis of Lansdowne.  
 3. It is famous for holding three meetings before Independence of India including the Cabinet Mission.  
*Select the correct answer using the codes given below.*  
 (a) 1 and 2 (b) 2 and 3  
 (c) 1 and 3 (d) All of these
- 2.** Which one among the following statements regarding Jorwe Culture is not correct?  
 (a) Pravara-Godavari Valley was the nuclear zone of Jorwe Culture  
 (b) The main sites of Jorwe Culture are Daimabad, Inamgaon, Jorwe and Nevasa  
 (c) At Daimabad, one gets archaeological evidence of symbolic burial  
 (d) Practically all settlement in the Northern Deccan were suddenly deserted

### 2012 (II)

- 3.** The first Indian ruler to be shown in images as wearing a dress akin to trousers is  
 (a) Chandragupta (b) Ashoka  
 (c) Kanishka (d) Samudragupta

- 4.** Who among the following cultures were the first to paint their pottery?  
 (a) Mesolithic (b) Chalcolithic  
 (c) Neolithic (d) Iron age

### 2013 (I)

**Directions** (Q. Nos. 5-6) *The following questions consist of two statements, Statement I and Statement II. You are to examine these two statements carefully and select the answers to these questions using the codes given below.*

#### Codes

- (a) Both the statements are true and Statement II is the correct explanation of Statement I  
 (b) Both the statements are true, but Statement II is not the correct explanation of Statement I  
 (c) Statement I is true, but Statement II is false  
 (d) Statement I is false, but Statement II is true

- 5. Statement I** Lomash Rishi and Sudama caves in the Barabar hills, modelled on wooden architectural prototypes are examples of the earliest cave architecture in India.

**Statement II** Barabar hills caves were dedicated by Chandragupta Maurya to Ajivika monks.

- 6. Statement I** Sculptures of the Gandhara School stylistically and typically linked to the Greco-Roman and the Parthian art of Iran.

**Statement II** The earliest stone Buddha images in the Swat valley pre-dated the Kushana period,

which suggests that certain iconographic conventions were already well-established in the pre-Kushana period.

- 7.** The highly polished monolithic Ashokan Pillars were carved out of single pieces of a buff-coloured sandstone, usually mined from the quarries of  
 (a) Chunar near Mirzapur  
 (b) Lauriya in Nandangarh  
 (c) Sarnath near Varanasi  
 (d) Udayagiri near Bhubaneswar

### 2014 (I)

- 8.** Which one of the following was a temple built by the Chola Kings?  
 (a) Brihadiswara Temple, Tanjavur  
 (b) Meenakshi Temple, Madurai  
 (c) Srirangam Temple, Thiruchirappalli  
 (d) Durga Temple, Aihole
- 9.** The University of Nalanda was set-up by which Gupta ruler?  
 (a) Kumaragupta II (b) Kumaragupta I  
 (c) Chandragupta II (d) Samudragupta

#### 10. Match the following

| List I (Temples)     | List II (Towns) |
|----------------------|-----------------|
| A. Kailasanatha      | 1. Bhubaneswar  |
| B. Lingaraj          | 2. Khajuraho    |
| C. Kandariya Mahadev | 3. Mount Abu    |
| D. Dilwara           | 4. Kanchipuram  |

#### Codes

| A B C D     | A B C D     |
|-------------|-------------|
| (a) 4 2 1 3 | (b) 4 1 2 3 |
| (c) 3 1 2 4 | (d) 3 2 1 4 |

[illegible]

# 02

# GEOGRAPHY

*The Geography section of the CDS examination plays an important role as around 12-14 questions are asked every year. It includes both Indian as well as world geography. Topics which are asked include Earth structure, Atmosphere, Rivers, Mountains, Oceanography, Latitude and Longitude, Location of important industries etc.*



## PART I PHYSICAL AND WORLD GEOGRAPHY

### THE UNIVERSE

- The Universe is commonly defined as the totality of everything that exists including all physical matter like the planets, stars, galaxies and the contents of intergalactic space and energy. The study of Universe is known as **Cosmology**.
- The Universe comprises of billions of galaxies. Galaxy is giant assemblies of stars, planets, gases and dust. These galaxies are made up of millions of stars held together by the force of gravity and these stars account for most of the masses of the galaxy. Andromeda is our nearest galaxy.
- Our own galaxy is called the **Milky Way** (the Akash Ganga) and it contains about 300 billion stars and one of these is our Sun. Planets and other objects go round the Sun and make up the solar system with the Sun at the centre.
- In AD 140, **Ptolemy** propounded the theory that the Earth was at the centre of the Universe and the Sun and the other heavenly bodies revolved around it.
- In 1543, **Copernicus** said that the Sun is at the centre of the Universe and not the Earth. Kepler supported Copernicus but said that the Sun is the centre of solar system and not the Universe.
- In 1924, **Edwin Hubble** first demonstrated the existence of galaxies beyond Milky Way.

*Structurally, the galaxies are found in three forms as follow*

- Spiral** have a central nucleus with great spiral arms. Milky Way and Andromeda are the examples of it.
- Elliptical** without spiral arms.
- Irregular** with no shape.

## Evolution of Universe

*The three main theories put forward to explain the origin and evolution of the Universe are as follows :*

- i. **Big Bang Theory** (Proposed by Georges Lemaitre) Big Bang was an explosion that occurred 13.8 billion years ago, leading to the formation of galaxies of stars and other heavenly bodies.
- ii. **Steady State Theory** This theory was proposed by Bondi, Gold and Fred Hoyle. According to this theory, new galaxies are continuously being formed and older ones are destroyed. The overall size of the Universe remains constant.
- iii. **The Pulsating Theory** According to this theory, the Universe is supposed to be expanding and contracting alternately i.e. pulsating. At present, the Universe is expanding.

## SOLAR SYSTEM

- The solar system comprises the Sun and 8 planets and their moons and other non-stellar objects, which are believed to have been developed from the condensation of gases and other lesser bodies. The size of solar system has been estimated about  $10^5$  Astronomical Unit (AU).
- The Sun is at the centre of the solar system and all the planets revolve around it in elliptical orbit. It is the nearest star to the Earth.
- It also consists of Interstellar debris such as asteroids, meteoroids, comets, electrically charged gases, called **Plasma** and interplanetary dust particles. The components of solar system other than planets, dwarf planets and satellites are called as Small Solar System Bodies (SSSB).

## THE SUN

- The Sun accounts for more than 99% of the mass of the solar system and due to this, the Sun exerts immense gravitational pull to keep the planets rotating around it in definite elliptical orbits.
- The Sun is the major source of energy in the solar system. The energy is provided by the nuclear fusion reaction, that converts hydrogen into helium in the core of the Sun.
- Super-imposed on Sun's white light are hundred of dark lines called **Fraunhofer** lines. Each line indicates some elements present in the solar system.

## Concepts Associated with the Sun

- **Solar Winds** The Sun is continuously emitting streams of proton in all directions either as spiral streams called **Solar Wind** or bouts of incandescent material called **Solar Flares**. Solar flares being hot ionised gases pose danger to satellite communication.

- **Aurora** The constituent particles of the solar wind are trapped by the Earth's magnetic field and enter the Earth's upper atmosphere as Aurora. It is described as Aurora Borealis in the Northern hemisphere and Aurora Australis in Southern hemisphere.
- **Plages and Sunspots** The surface of the Sun is continuously changing. Bright spots are called **Plages** and dark spots are called **Sunspots**. The Sunspots are cold and dark regions on the Sun's surface with a periodicity of 11 years. These spots greatly influence the global climate.

## Planets

A planet is a heavenly body that moves in an orbit around a star, such as the Sun. There are 8 planets in solar system. These planets move around the Sun in the fixed path.

### Mercury

- It is the innermost and smallest planet in the solar system.
- Mercury is nearest to the Sun, having **rotation period** is 58.65 days and **revolution period** is of 88 days (**fastest in solar system**) with no satellite, it has the highest diurnal range of temperature and has the shortest year.

### Venus

- Also known as **Earth's twin**, **morning star** and **evening star**. Brightest heavenly body after the Sun and the Moon because of 70% albedo. Venus is closest planet to Earth.
- It is **hottest planet** in our solar system because of 97%  $\text{CO}_2$ . It is slightly smaller than the Earth (500 km less in diameter). It rotates clockwise (East to West) and it does not have satellite.
- It has **slowest rotation** period in our solar system (257 days) and have almost equal **rotation** and **revolution** (224.7 days) period.

### Mars

- Mars is the fourth nearest planet to the Sun. It is also known as **Red Planet** because it appears as a reddish ball when looked through a telescope.
- Its **revolution period** is 687 days and **rotation period** is of 24.6 hrs (almost equal to that of Earth), it has two satellites **Phobos** and **Deimos** and it has signs of water, hence, there a possibility of life.
- Various space missions have been sent e.g. Vikings, Pathfinder, Mars Odyssey, Mars Orbiter Mission.

## Jupiter

- Largest of all planets, almost 11 times of the Earth, it is also called **Lord of the Heavens**.
- It has fastest rotational time in our solar system (9.8 hrs) and **revolution period** is of 12 years, it has 16 satellites (prominent are Europa, Ganymede and Callisto).
- **Europa** resembles the living condition like that of the Earth. **Ganymede** is the largest satellite of our Solar system.

## Saturn

- Second largest planet (in size) after Jupiter and **Least dense** of all (30 times less dense than the Earth). It has **revolution period** of 29 years and **rotation period** is of 10.3 hours.
- It has around 62 satellites. Titan is the most prominent one. Its atmosphere is dominated by presence of nitrogen. It has a system of three well defined rings.

## Uranus

- Identified as a planet in 1781 by William Herschel. It rotates from **North to South** as it is inclined at an angle of 98° to its orbit, and it has **revolution period** of 84 years and **rotation period** is of 10.8 years.
- Like Saturn, it is also surrounded by a system of 5 faint rings and it has 21 satellites (Miranda, Ariel etc.)

## Neptune

- Appears as 'Greenish Star' because of presence of 'Methane'. It has **revolution period** of 165 years and **rotation period** is of 15.7 days.
- Discovered by **JG Galle of Berlin** in 1846. Its prominent satellite are 'Triton and Nereid'. Uranus and Neptune are called Jovian Twins.



## PLUTO

Pluto was discovered in 1930 and was considered as ninth planet in our solar system. However, it is no longer considered a planet now. It orbits beyond the orbit of Neptune.

It is much smaller than any of the official planets and now classified as a **dwarf planet**. It is the second largest dwarf planet in the solar system.

## Facts Related to Planets

- Biggest planet – *Jupiter*
- Smallest planet – *Mercury*
- Biggest satellite – *Ganymede*
- Smallest satellite – *Deimos*
- Blue planet – *Earth*
- Green planet – *Uranus*
- Brightest planet – *Venus*
- Brightest star – *Sirius (Dog Star)*
- Closest Star of solar system – *Proxima Centauri*
- Coldest planet – *Neptune*
- Evening star – *Venus*
- Red planet – *Mars*
- Slowest revolution in solar system – *Neptune*
- Slowest rotation in solar system – *Venus*
- Earth's twin – *Venus*
- Only satellite with an atmosphere like Earth – *Titan*

## Comet

- A comet is a member of the Sun's family, part of the solar system. Comet travels on a path or elliptical orbit around the Sun on a regular schedule.
- It may have originated from a huge cloud the 'oort cloud' that is thought to surround the solar system.
- It has a head and a tail. Its tail originates once it gets closer to the Sun. The brightest part of the comet is head (coma) and it is made up of solid matter combined with gases.

## Asteroids

- Asteroids or minor planets circle in a broad belt between the orbits of **Mars** and **Jupiter**.
- It is the debris left over from the formation of inner planets, also called '**planetoids**' or **small planets** and they are chunks of rock covered in frozen gases.

## Meteoroids, Meteors and Meteorites

- Meteoroids are small bodies that travel through space. Meteoroids are smaller than Asteroids, most are smaller than the size of a pebble.
- A meteor is a meteoroid that has entered the Earth's atmosphere, usually making a fiery trail as it falls. It is sometimes called a **shooting star** or a **falling star**.
- A meteor shower is a phenomenon in which many meteors fall through the atmosphere in a relatively short time and in approximately parallel trajectories. A very intense meteor shower is called a **Meteor storm**.
- A Meteorite is a meteor that has fallen to Earth. Meteorites are made up of rock and/or metals.

## The Moon

- Moon is the only natural satellite of Earth. It has no atmosphere as due to weak gravitational force the gases easily escape from its surface.
- Only 59% of Moon surface is directly visible from Earth. It takes 27 days, 7 hrs, 43 min and 11.47 sec to complete one revolution around the Earth. It rotates on its axis in exactly the same time as that in revolution. That is why we see only one side of the Moon.

## Phases of the Moon

- Due to its spherical shape, only one-half of the Moon gets illuminated by the Sun.
- Visible surface of Moon is not same everyday because it revolves round the Earth and visible shape of the Moon are referred to as the phases of the Moon.
  - **New Moon** This is the phase when the Moon is between the Earth and the Sun and consequently the part of the Moon facing us is in complete darkness. (New Moon = No Moon)